AI MENTAL HEALTH UK

Task - Baseline stats model

Duration – 1-3 May

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Domain - Data Science

Objective

Build statistical model of MH prevalence rates by region. Apply confidence intervals.

To model regional prevalence of common mental disorders (CMD) using APMS 2014 data and quantify uncertainty using confidence intervals.

Data Source

- CMD prevalence: apms-2014-ch-02-tabs - **Table 2.11 (Age-standardised, All adults, Any CMD)**
- Sample size: apms-2014-ch-14-tabs - **Table 14.2**

	Region	CMD_Prevalence	n	p
0	North East	15.7	677	0.157
1	North West	19	1785	0.19
2	East Midlands	16.7	1200	0.167
3	West Midlands	18.4	1419	0.184
4	East of England	14.4	1367	0.144
5	London	18	1530	0.18
6	South East	13.6	756	0.136

Method

- Age-standardised values used to ensure comparability
- Prevalence converted to proportions
- Wilson confidence intervals computed using regional sample sizes

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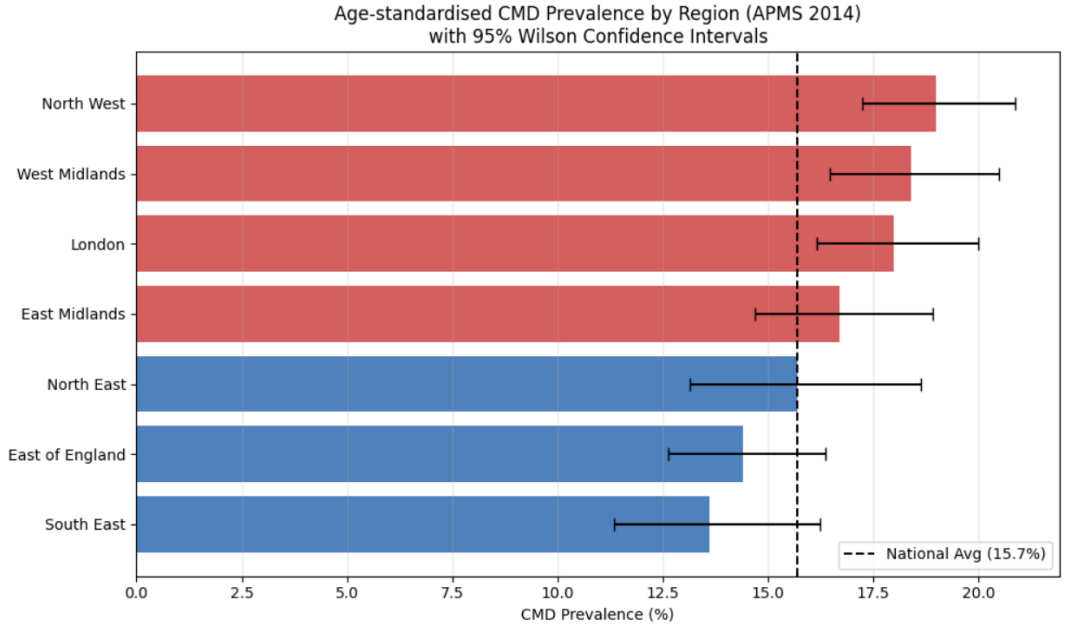
Results

- Regions such as **South West and West Midlands show higher CMD prevalence**
- Regions like **South East show lower prevalence**
- Smaller sample sizes produce wider confidence intervals



Conclusion

The model highlights meaningful regional variation in CMD prevalence while accounting for uncertainty, providing a statistically robust baseline.